

Course Code: 105-01
Course Title: Introduction to Cybersecurity and Information Security

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Course Title	Introduction to Cybersecurity and Information Security
Credits	4
Course Category	Major Course
Level of Course	200-299 (Intermediate Level)
Teaching per Week	4 Hrs.
Minimum weeks per Semester	15 (Including class work, examination, preparation etc.)
Review / Revision	-
Implementation Year:	A.Y. 2024-2025
Purpose of Course	The purpose of the "Introduction to Cybersecurity and Information Security" course is to provide students with a foundational understanding of the key principles and practices in the field of cybersecurity. This course aims to equip students with knowledge of various types of cyber threats and vulnerabilities, and the strategies used to protect information systems. By introducing essential concepts such as cryptography, network security, and information security management, the course prepares students to recognize and mitigate security risks. It also covers emerging trends and technologies in cybersecurity, ensuring students are aware of current and future challenges. Additionally, the course offers insights into career paths in cybersecurity, guiding students on the skills and certifications needed for professional growth in this critical and evolving domain. Overall, the course aims to build a solid base for further study and a successful career in cybersecurity and information security.
Course Objective	1. Understand Fundamental Concepts: To provide students with a thorough understanding of the basic principles and terminology of cybersecurity and information security, including the CIA triad (confidentiality, integrity, availability). 2. Identify and Analyze Cyber Threats: To enable students to identify various types of cyber threats and vulnerabilities, and analyze real-world cyber incidents to understand their impact on information systems. 3. Learn Cryptographic Techniques: To introduce students to cryptographic principles, algorithms, and protocols, and to demonstrate how these techniques are used to protect data confidentiality and integrity. 4. Gain Practical Network Security Skills: To equip students with the knowledge and skills required to secure networks, including the use of firewalls, intrusion detection systems (IDS), and securing wireless networks. 5. Develop Security Management Strategies: To teach students how to develop and implement effective information security policies and procedures, manage security risks, and ensure compliance with legal and regulatory requirements.
Pre-requisite	Concepts of computer data and necessity of keeping the computers secure.
Course Outcomes	CO1: Comprehensive Understanding of Cybersecurity Principles: Students will be able to explain the fundamental concepts of cybersecurity and information security, including the CIA triad, and understand the importance of these principles in protecting information systems. CO2: Identification and Mitigation of Cyber Threats: Students will be capable of identifying various types of cyber threats and vulnerabilities, analyzing real-world cyber incidents, and applying appropriate strategies to mitigate these threats. CO3: Proficiency in Cryptographic Methods : Students will demonstrate

	proficiency in using cryptographic algorithms and protocols to secure data, ensuring its confidentiality, integrity, and authenticity. CO4: Practical Network Security Skills : Students will acquire practical skills in securing network environments, including configuring and managing firewalls, implementing intrusion detection systems (IDS), and securing wireless networks against common threats. CO5: Development of Security Policies and Risk Management: Students will be able to develop and implement effective information security policies and procedures, conduct risk assessments, and ensure compliance with relevant legal and regulatory standards, enhancing the overall security posture of an organization.								
Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO)		PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PSO7	PSO8
	CO1								
	CO2								
	CO3								
	CO4								
	CO5								
Course Content	Unit 1: Fundamentals of Cybersecurity and Information Security 1.1 Introduction to Cybersecurity 1.1.1 Definition and importance of cybersecurity 1.1.2 History and evolution of cybersecurity 1.1.3 Key concepts: confidentiality, integrity, availability (CIA Triad) 1.2 Types of Cyber Threats 1.2.1 Malware (viruses, worms, trojans) 1.2.2 Phishing, social engineering 1.2.3 Denial of Service (DoS) and Distributed Denial of Service (DDoS) 1.3. Cyber Attack Vectors 1.3.1 Network-based attacks 1.3.2 Host-based attacks 1.3.3 Application-based attacks Unit 2: Basic Cryptography 2.1 Introduction to Cryptography 2.1.1 Purpose and applications of cryptography 2.1.2 Symmetric vs. asymmetric cryptography 2.2 Cryptographic Algorithms 2.2.1 Symmetric key algorithms: DES, AES 2.2.1 Asymmetric key algorithms: RSA, ECC 2.3 Cryptographic Protocols 2.3.1 Public Key Infrastructure (PKI) 2.3.2 Digital signatures 2.3.3 Hash functions and integrity Unit 3: Network Security Basics 3.1 Network Fundamentals 3.1.1 Overview of network architecture and protocols 3.1.2 OSI and TCP/IP models 3.2 Network Security Devices 3.2.1 Firewalls: types and configurations 3.2.2 Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) 3.3 Securing Wireless Networks 3.3.1 Wi-Fi security protocols: WEP, WPA, WPA2 3.3.2 Best practices for securing wireless networks								

	Unit 4: Information Security Management 4.1 Information Security Principles 4.1.1 Information Security Management Systems (ISMS) 4.1.2 Risk management and assessment 4.2 Security Policies and Procedures 4.2.1 Developing and implementing security policies 4.2.2 Incident response planning and management 4.3 Compliance and Legal Issues 4.3.1 Overview of key regulations (GDPR, HIPAA, etc.) 4.3.2 Ethical considerations in cybersecurity Unit 5: Emerging Trends and Careers in Cybersecurity 5.1 Current Trends in Cybersecurity 5.1.1 Cloud security 5.1.2 Internet of Things (IoT) security 5.2 Future of Cybersecurity 5.2.1 AI and machine learning in cybersecurity 5.2.2 Quantum computing and its impact on security 5.3 Career Paths in Cybersecurity 5.3.1 Roles and responsibilities of cybersecurity professionals
Reference Books	1. "Cryptography and Network Security", Author: Atul Kahate, Publisher: McGraw Hill Education, ISBN: 978-9332901803 2. "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", Author: Nina Godbole, Sunit Belapure, Publisher: Wiley India, ISBN: 978-8126521798 3. "Network Security and Cryptography", Author: Bernard L. Menezes, Publisher: Cengage Learning India, ISBN: 978-8131500720. 4. "Information Security: Principles and Practice", Author: Mark Stamp, Publisher: Wiley India, ISBN: 978-8126563965. 5. "Computer Security: Principles and Practice", Author: William Stallings, Lawrie Brown, Publisher: Pearson India, ISBN: 978-9332585225. 6. "Cybersecurity Essentials", Author: Charles J. Brooks, Christopher Grow, Philip Craig, Donald Short, Publisher: Wiley India, ISBN: 978-8126579706 7. "Fundamentals of Cyber Security", Author: Mayank Bhushan, Rajkumar Singh Rathore, Aatif Jamshed, Publisher: BPB Publications, ISBN: 978-9386551551 8. "Ethical Hacking", Author: K. V. Naveen, Publisher: Notion Press, ISBN: 978-9352065092. 9. "Information Security: Principles and Practices", Author: Dhiren R. Patel, Publisher: PHI Learning Pvt. Ltd., ISBN: 978-8120343259 10. "Introduction to Information Security and Cyber Laws", Author: Surya Prakash Tripathi, Ritendra Goel, Praveen Kumar Shukla, Publisher: Dreamtech Press, ISBN: 978-9351194731 11. "Cyber Security: Threats and Responses for Government and Business", Author: Jack Caravelli, Nigel Jones, Publisher: Jaico Publishing House, ISBN: 978-8184958696 12. "Computer and Information Security Handbook", Author: John R. Vacca, Publisher: Elsevier India, ISBN: 978-8131221564.
Teaching Methodology	Class Work, Discussion, Lab work, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.